



AI in Software Testing

Exploring AI in software testing, where automation and precision reign supreme.

Software testing is an indispensable phase in the software development lifecycle, ensuring that applications perform as expected and are free from defects that could affect user experience or system functionality. AI's role in software testing is becoming increasingly significant, offering novel ways to automate and enhance testing processes. This chapter builds upon the previous discussions of AI in coding, focusing on how AI is reshaping software testing.

Software Testing in the Era of AI

AI is transforming traditional testing practices by introducing automation, predictive analytics, and advanced anomaly detection. AI-powered testing tools can analyze application data, learn from past test cases, and predict where future failures are likely to occur.

Infusion of AI is enabling broader and more comprehensive test coverage by quickly generating test cases that cover edge cases and scenarios which might be missed by manual testing.

AI-Powered Software Testing Tools

AI-powered software testing tools are redefining the boundaries of quality assurance by bringing in unprecedented levels of efficiency, accuracy, and coverage. These tools use machine learning, pattern recognition, and intelligent algorithms to automate complex testing processes. Let's take a closer look at how these tools function and how they can be implemented in software testing workflows.

1) Selenium with AI

Selenium is a robust tool for automating web browsers, which